

Security risk assessment report

The organization recently experienced a major data breach, which compromised the safety of their customers' personal information, such as names and addresses. Your organization wants to implement strong network hardening practices that can be performed consistently to prevent attacks and breaches in the future.

After inspecting the organization's network, you discover four major vulnerabilities. The four vulnerabilities are as follows:

The organization's employees share passwords.

The admin password for the database is set to the default.

The firewalls do not have rules in place to filter traffic coming in and out of the network.

Multifactor authentication (MFA) is not used.

If no action is taken to address these vulnerabilities, the organization is at risk of experiencing another data breach or other attacks in the future.

Part 1: Select up to three hardening tools and methods to implement

- Require users to verify their identity using an additional authentication factor such as MFA or a security token.
- Require unique passwords for each user account, prohibit password sharing, and change the database administrator account from the default password to a strong, unique password.
- Implement firewall rules to monitor and control inbound and outbound network traffic, allowing only authorized communications and blocking unauthorized access attempts.

Part 2: Explain your recommendations

The organization faces several security risks that could result in unauthorized access and data breaches. Employees should be prohibited from sharing

passwords, and strong, unique passwords should be required for all accounts to improve accountability and access control. The default database administrator password should be replaced with a strong, unique password to prevent unauthorized access. Firewall rules should be configured to filter incoming and outgoing network traffic and reduce exposure to cyber threats. Additionally, implementing multifactor authentication (MFA) will provide an extra layer of security and strengthen the organization's overall security posture.